

Suggested outline (From Amanda):

1. Introduction
 - a. Overview of the major issue you're looking at and why it's important
 - b. Literature review: data, methods, etc.
2. Methods
 - a. What are the statistical methods you are focusing on? Modeling approach, estimation technique, evaluation method, etc?
3. Simulation Study
 - a. Results from applying the method to simulated data and discussion of when it works/doesn't work
4. Results
 - a. Apply the method to a real dataset and interpret the results
5. Discussion

Paper outline -

Abstract

Introduction

- Major issue we're looking at: examiner decision making in forensic studies
 - These have implications for jury decision making and outcomes of criminal cases
- Introduction to our Bayesian approach as well as that we are using ordinal probit model
- Talk about the variables we discussed using and which ones we landed on
- Mention that we are using cartridge case data as opposed to bullet/fingerprint
- Touch on likelihood ratios
- Brief description of Bayesian

Literature Review

The following pieces of literature gave us insight into how we approached modeling through a Bayesian lens. We used analysis of casing likelihood ratios put forth by forensic examiner Max Gyll as a baseline for our analysis. There are also valuable contributions to Gyll's analysis made by Michael Rosenblum et. al, in their response. We landed on adapting this analysis to an ordered probit model using concepts put forth by Thomas Busey for fingerprint analysis. Finally, our data as well as additional theoretical framework came from Keith Monson.

Previous Work

Guyll used Bayesian processes to calculate the posterior odds that a cartridge case from a crime scene was fired from the same reference firearm as the “same source” or “different source” given the decision made by a trained forensics examiner. The prior odds of the equation is represented by previously assumed odds of same source vs. different source date while the likelihood ratio is defined as the examiner’s decision given whether the casing is from the same source or a different source.

A response to the Guyll article made by Michael Rosenblum et. al finds issues in his statistical reasoning. Specifically relating to how Guyll’s calculated likelihood ratios produced inflated data on how often cartridge cases were fired from a reference firearm. The qualm with this methodology is that it is very much skewed in favor of the prosecution as opposed to the defendant. Given Guyll’s methodology, there is an issue with the prior that he devised. The prior odds in the case of Guyll are equivalent to one, which imply a 50% probability to the reference firearm and a 50% probability to be distributed across all other potential firearms that could, in theory, have been used. This inherently favors the reference firearm, which is indicative to inflated odds that it will be the conclusion. This type of bias is referred to as equiprobability bias, which is defined as when individuals incorrectly believe that all possible outcomes in a random process are equally likely to occur, when, in reality they are not. Rosenblum puts forth the argument that changing these prior assumptions. They use a $1/n$ prior as an illustration showing how the posterior probability rapidly decreases as the number of potential firearms (n) increases.

Methodology Framework

Busey -

Thomas Busey et al. used data from fingerprint error rate studies to first estimate distributions of examiner decisions, then created a likelihood ratio to measure the strength of support for same-source and different-source pairs. They used this likelihood ratio multiplied by the prior odds to get the posterior odds, which represent how much more guilty than innocent the suspect is. This process assumes that evidence used is probative and credible, motivating our decision to drop pairs categorized as unsuitable in our analysis. In order to estimate the latent distribution of decisions per gun, they applied an ordered probit model, using a 5-level conclusion scale (Elimination, Inconclusive-C, Inconclusive-B, Inconclusive-A, Identification) as the ordered categories. These categories are ordered such that an elimination decision provides the least amount of evidence for a same-source proposition and an identification decision provides the most. The ordered probit model assumes that a latent, unobserved variable exists. In the forensics case, examiners end up somewhere along this latent axis after making decisions. While we can’t measure exact values of their endpoints on this latent axis, their position on the axis represents their strength of support for either a different-source proportion or same-source proposition. Additionally, decision criteria exist such that examiners above a certain threshold on the latent scale change their decision: from Inconclusive-A to Identification, for example. These thresholds exist globally, not individually per examiner. Finally, the ordered probit model assumes that

latent values across examiners can be modeled by a normal distribution. The proportions of examiner responses for a given decision can be obtained by the area under the normal curve that represents that decision's threshold. The mean of the normal distribution represents the support for either different or same-source propositions, with larger values of μ supporting same-source decisions more. The standard deviation (σ) of the normal distribution represents consistency across examiners. Low standard deviation indicates that examiners tend to make the same decisions regarding a certain pair, and high standard deviation means that examiners don't tend to agree.

Data

In their 2022 study, "Accuracy of comparison decisions by forensic firearms examiners", Kieth Monson et. al employed a "Black Box" experimental design to test the accuracy of 173 qualified U.S. forensic examiners. These examiners were asked to complete 8640 total comparisons of firearms spanning from both bullets and cartridge casings. Examiners were provided sets of evidence and asked to determine whether they were fired from the same firearm, a different firearm, or whether the result was inconclusive. Overall, the examiners were very accurate in their findings, with correct rates that were well above 90%. However, when errors were made, it is noted that these errors are concentrated within a subset of examiners. Additionally, there was the option of classifying decisions as inconclusive, which speaks to the varying range of difficulties from one comparison to others. The results from Monson et. al suggest relatively high accuracy in the comparison of firearms, and additionally suggest that those examiners who are more experienced and generally have higher qualifications are less prone to Type I and Type II errors.

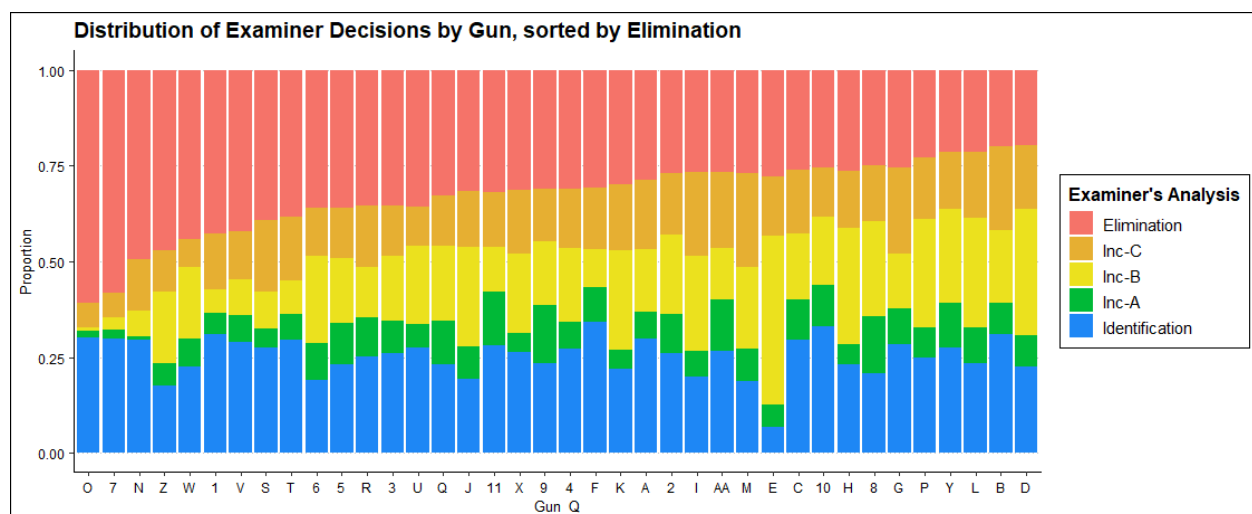
The data that we ran our simulations on comes from Kieth Monson et al.'s paper "Accuracy of comparison decisions by forensic firearms examiners". The data consists of decisions made by firearm examiners on both bullets and casings. We decided to exclusively focus on the casings part of the dataset in order to keep consistent with the analysis made by Gyll about likelihood ratios. Additionally, this dataset has a comprehensive explanation of the different variables that are used. There are 4320 comparisons made between bullet casings as well as 24 different explanatory variables. After filtering out the variables we saw as redundant as well as the ones that were unnecessary for our analysis, we were left with only seven of the original 24. These seven are ExamID (Three letter code assigned to examiner for packet analysis. First two letters are unique to a particular examiner and remain the same for the examiner for all mailing rounds), Examiner's Analysis (Categorical: (ID, Inc-A,B, C, Elim, Unsuitable), Gun_1 (Letters indicate samples from Beretta gun. Numbers indicate samples from NON_Beretta gun), Gun_Q (Letters indicate samples from Beretta gun Numbers indicate samples from NON_Beretta gun.), M_NM (Either Match or nonmatch), ExamID_2 (These two letters are unique to a particular examiner and remain the same for the examiner for all mailing rounds) and

Pair (Letters indicate samples from Beretta gun. Numbers indicate samples from NON_Beretta gun). In terms of the different choices, ID is defined as the gun being identified as a match, INC-A is the examiner classifying the comparison as inconclusive, but leaning towards identification. INC-B is tire inconclusiveness that does not favor identification or elimination. INC-C is inconclusiveness that favors elimination and elimination is the gun being eliminated as a match. We were going to use the data proposed by Gyll, however, we ran into an issue as there was no information to indicate where each gun came from.

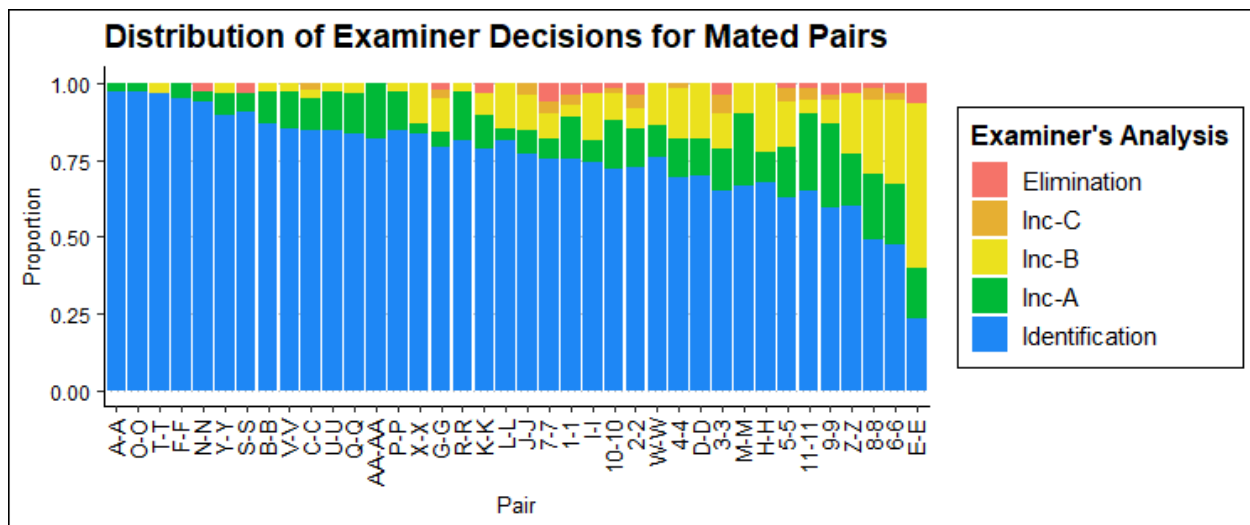
- Other non-firearm? - Maybe...only if it is very justified

EDA

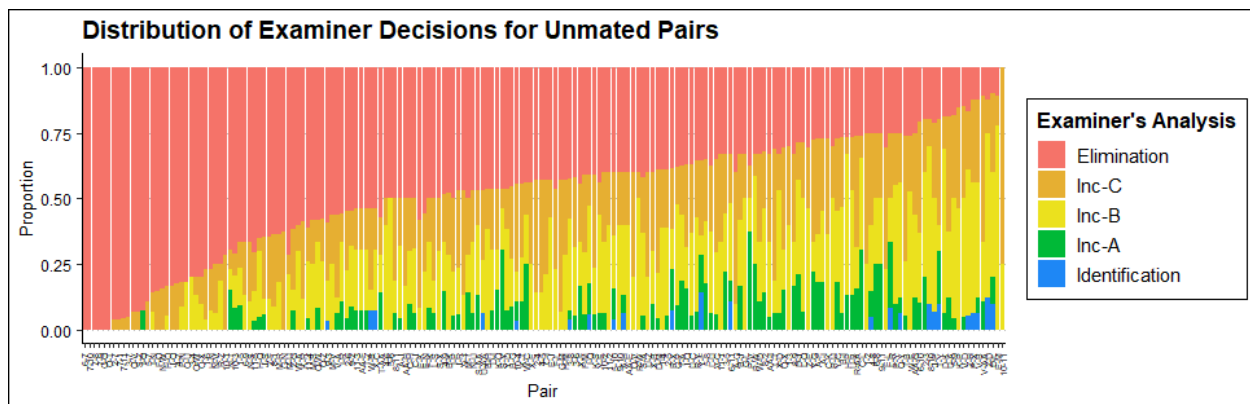
The jitter plot above shows the frequency with which decisions were made in each category for both mated and nonmated comparisons. The identification and elimination frequencies align with our initial expectations. However, we observe some discrepancies in the results for INC-B and INC-A. Because INC-A represents an inconclusive result that leans toward identification, we would expect more points to be concentrated in the mated portion of the plot. Instead, the plot shows relatively even numbers of points in both the mated and nonmated sections. A similar discrepancy appears for INC-B. Since INC-B is intended to represent a completely neutral inconclusive decision, we would expect roughly equal numbers of points in the mated and nonmated sections. However, the plot shows substantially greater representation in the nonmated category compared to the mated category.



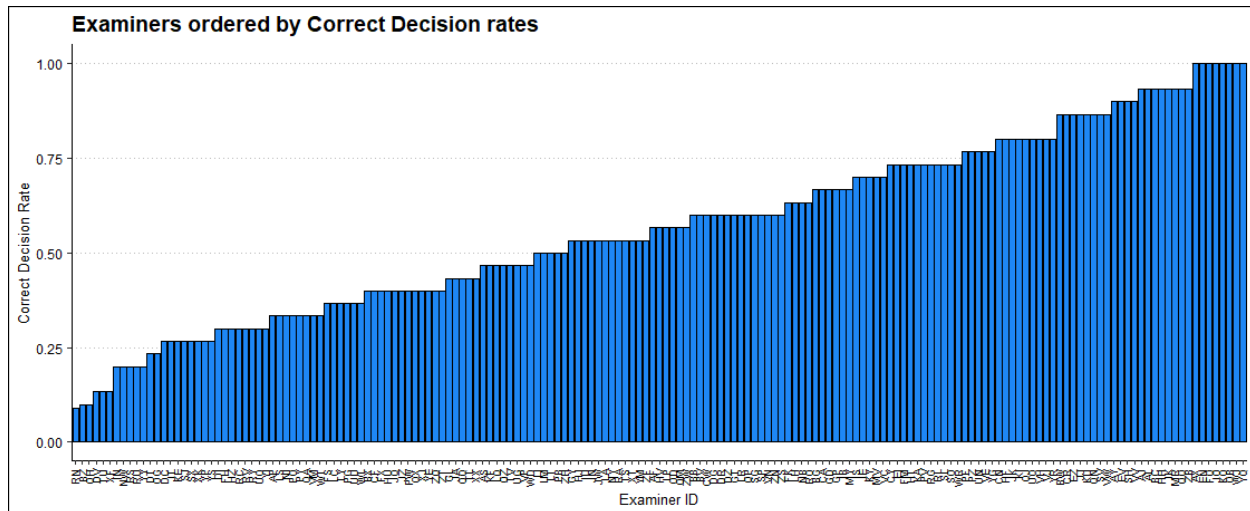
- Note about this plot: how informative even is this plot? Because guns can be involved in pairs that are both unmated and mated, this doesn't really tell us anything about the decisions made on them.



- This plot shows that for some mated gun pairs, A-A for example, examiners seemed to have an easier time making the correct decision. We see almost all identifications made on this pair, with a very small percentage of decisions being the second highest support for same-source, Inconclusive-A. On the contrary, some pairs seem more difficult to identify as the same-source. E-E has a very low identification rate and a very high Inconclusive-B rate, showing ambiguity in examiner's responses.



- How can we get this graph to better show the pairs, given there are so many unmated pairs? Similar analysis to graphs of mated pairs. Some pairs, 6-7 for example, have 10% elimination rates, and some on the opposite end of the spectrum have very low to even 0%.



- Examiners ordered by correct decision rates. Some examiners have 100% correct rates, and some have less than 10%. A decision is categorized as ‘correct’ if, for a nonmated pair, an examiner concluded “Elimination” or, for a mated pair, an examiner concluded “Identification.” Inconclusive decisions on any level were not counted as correct.

Methodology

- Use of the ordinal probit model
- How we are calculating likelihood ratios
- Recreation of Busey et al.

Simulation Study

- MCMC sims for latent dists, likelihood ratio results

Results

- Using latent distributions on new, synthetic dataset
- Comparison of accuracy of our model to original data

Discussion

- Prob easy to expand on the study by mentioning finger print and bullets
- Shortcomings of our analysis
- Real-world implications
 - Black box studies
 - Court cases/results, jury priors and decisions
- Future directions